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Object-oriented programming (OOP) is a programming paradigm that revolves around the concept of objects. An object is an instance of a class, which is a blueprint or template for creating objects. Here are some key OOP concepts:

Encapsulation: Encapsulation is the process of hiding the implementation details of an object from the outside world. This is achieved by making the data and methods of an object private, so that they cannot be accessed directly.

Inheritance: Inheritance is the process of creating a new class from an existing class. The new class inherits the properties and methods of the existing class and can add new functionality or override existing functionality.

Polymorphism: Polymorphism is the ability of objects of different classes to be used interchangeably. This is achieved through inheritance and interfaces, where different classes can implement the same methods in different ways.

Abstraction: Abstraction is the process of simplifying complex systems by focusing on the most important details. In OOP, this is achieved by creating abstract classes or interfaces that define a set of methods without specifying their implementation.

Object: An object is an instance of a class. It has state (represented by its attributes) and behavior (represented by its methods).

Class: A class is a blueprint or template for creating objects. It defines the attributes and methods of the objects that will be created

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SDLC stands for Software Development Life Cycle. It is a process that software development teams use to plan, design, develop, test, and maintain software. The SDLC model provides a framework for the software development process, which typically consists of the following stages:

Requirements gathering: This stage involves understanding the customer's needs and defining the features and functionality that the software must have.

Design: In this stage, the software architecture is designed, including the software components and their relationships.

Development: The actual coding and implementation of the software takes place in this stage.

Testing: The software is tested for functionality, performance, and reliability in this stage.

Deployment: The software is released to the customer or end-user in this stage.

Maintenance: After the software has been deployed, any issues that arise are addressed, and the software is updated or improved as needed.

There are various SDLC models, including the Waterfall model, Agile model, Spiral model, and V-Model. Each model has its advantages and disadvantages and is suitable for different types of projects.